Supplemental Materials: Math Glossary

1 Structure

1.1 Monoids and Groups

Definition 1.1. A monoid is a set with an associative binary operator and an identity element.

Definition 1.2. A group is a monoid where every element is invertible.

1.2 Actions

Since our proposed methods work well for a range of possible settings, we abstract the notion of data type into some object in a category with some monoid action on the object. Here, we deliberately refrain from stating the structure on the data - e.g. group, vector space, partial order - just that there is some category the data comes from. Here, by a monoid action of a monoid M on an object X, we mean a morphism $\mathtt{act} : G \times X \to X$:

Definition 1.3. An **action** is a morphism $\mathtt{act} : G \times X \to X$. An action has the properties of identity $\mathtt{act}(e, x) = x$ for all $x \in X$ and associativity $\mathtt{act}(g, \mathtt{act}(f, x)) = \mathtt{act}(f \circ g, x)$ for $f, g \in G$.[7]

Elements of X can be from one data field or all of them or some subset; similarly the actions act on the elements of X and each action can be a composition of actions.

1.3 Topology

There are numerous equivalent ways to formally define a topology. For our work, it is most convenient to use the *neighborhood* axioms of a topology, where a set X is a topological space if it is equipped with a function $N : X \to 2^{2^X}$ that assigns to any $x \in X$ the non-empty collection of all *neighborhoods* of x. The exact function chosen determines what *neighborhood* means for a given topological space.

Definition 1.4. The function N is called a *neighborhood* topology if the following axioms are satisfied; and then X with N is called a topological space. The following axioms must hold for each x in X:[5]

1. if N is a neighborhood $N \in \mathcal{N}(x)$ of x then $x \in N$
2. every superset of a neighborhood of x is a neighborhood of x; therefore a union of a neighborhood and adjacent points in X is also a neighborhood of x
3. the intersection of any two neighborhoods of x is a neighborhood of x
4. any neighborhood N of x contains a neighborhood $M \subset N$ of x such that N is a neighborhood of each of the points in M

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2 Category Theory

2.1 Categories

Definition 2.1. An **category** $\mathcal{C}$ consists of the following *data*:

1. a collection of *objects* $X \in \mathbf{ob}(\mathcal{C})$
2. for every pair of objects $X, Y \in \mathbf{ob}(\mathcal{C})$, a set of *morphisms* $X \xrightarrow{f} Y \in \mathrm{Hom}_{\mathcal{C}}(X, Y)$[1]
3. for every object X, a distinct *identity morphism* $X \xrightarrow{\mathrm{id}_x} X$ in $\mathrm{Hom}_{\mathcal{C}}(X, X)$
4. a *composition function* $f \in \mathrm{Hom}_{\mathcal{C}}(X, Y) \times g \in \mathrm{Hom}_{\mathcal{C}}(Y, Z) \to g \circ f \in \mathrm{Hom}_{\mathcal{C}}(X, Z)$

such that

1. *unitality:* for every morphism $X \xrightarrow{f} Y$, $f \circ \mathrm{id}_x = f = \mathrm{id}_y \circ f$
2. *associativity:* if any three morphisms f, g, h are composable,

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[rrr, bend right, dashed, "h\circ(g\circ f)=(h\circ g)\circ f"'] & Y \arrow[r, "g"] & Z \arrow[r, "h"] & W \end{tikzcd}$$

then they are associative $h \circ (g \circ f) = (h \circ g) \circ f$ [10, 14, 11, 6].

The composability property expresses that the morphism is transitive, while associativity expresses that the morphisms can be curried in various equivalent groupings. By formally specifying the properties of the topological structure data types as $\mathcal{K}$, we can express that these are the properties that are required as part of the implementation of the data type objects.

2.2 Functors

Functors are mappings between categories that preserve the domains, codomains, composition, and identities of the morphisms within the category[14],

Definition 2.2. [3, 4] A **functor** is a structure preserving map $F : \mathcal{C} \to \mathcal{D}$ between categories, which means it is a function between objects $F : \mathbf{ob}(\mathcal{C}) \mapsto \mathbf{ob}(\mathcal{D})$ and that for every morphism $f \in \mathrm{Hom}(C_1, C_2)$ there is a corresponding function $F : \mathrm{Hom}(C_1, C_2) \mapsto \mathrm{Hom}(F(C_1), F(C_2))$. A **functor** must satisfy the properties

- *identity*: $F(\mathrm{id}_C(C)) = \mathrm{id}_D(F(C))$
- *composition*: $F(g) \circ F(f) = F(g \circ f)$ for any composable morphisms $C_1 \xrightarrow{f} C_2$, $C_2 \xrightarrow{g} C_3$

$F(C) \in \mathbf{ob}(\mathcal{D})$ denotes the object to which an object C is mapped, and $F(f) \in \mathrm{Hom}(F_{(C_1)}, F_{(C_2)})$ denotes the morphism that f is mapped to.

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[1] Hom(source, target) denotes the set of maps from source to target

2.3 Natural transforms

Natural transforms are structure preserving maps of functors [12].

Definition 2.3. Given two functors $F, G\colon \mathcal{C} \to \mathcal{D}$, a **natural transformation** $\alpha : F \to G$ is a function which assigns to each object c of $\mathcal{C}$ a morphism $\alpha_c : F(c) \to G(c)$ of $\mathcal{D}$, in such a way that for every morphism $f : c \to c'$ in $\mathcal{C}$, the morphisms $F(f), G(f)$ in $\mathcal{D}$ commute such that $\alpha'_c(F(f)(F(c))) = G(f)(\alpha_c(F(c)))$. When this holds, α_c can be said to be *natural* in $\mathcal{C}$.[11].

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$\mathcal{C}$	$\mathcal{D}$
$\begin{tikzcd} c \arrow[d, "f"] \\ c' \end{tikzcd}$	$\begin{tikzcd} F(c) \arrow[r, "\alpha_c"] \arrow[d, "F(f)"] & G(c) \arrow[d, "G(f)"] \\ F(c') \arrow[r, "\alpha_{c'}"] & G(c') \end{tikzcd}$

Table 1: A natural transform from $F : C \to D$ to $G : C \to D$ is a family α_* of transformations for objects, so that $\alpha_c : F(c) \to G(c)$ is compatible with all the functor structures in the diagram above.

3 Structure Preservation

3.1 Homomorphism

Given the function $f : X \to Y$, with monoids $(X, \circ)$ and $(Y, *)$

Definition 3.1. A function f is a monoid **homomorphism** when $f(x_1 \circ x_2) = f(x_1) * f(x_2)$ and the function preserves identities $f(I_x) = I_y$ for all $x, y \in X$ [7]

which means that the operators $\circ$ and $*$ are compatible.

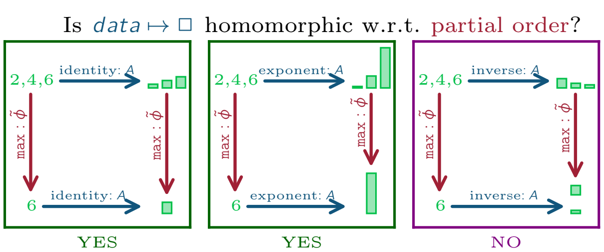

Figure 1: Encoding data as bar height (f) using an inverse transform $\circ$ is not homomorphic because the largest number is mapped to the smallest bar while the max function $(*)$returns the largest bar.

In Figure 1, the $\geqslant$ operator $(\circ)$ is defined as the compatible closed functions $\mathtt{max}(*)$ and the inverse transform is not homomorphic because it does not encode the maximum data value as the maximum bar value.

3.2 Equivariance

Given a group action G that acts on both X and Y

Definition 3.2. A function $f : X \to Y$ is **equivariant** when $f(act(g, x)) = act(g, f(x))$ for all g in G and for all x in X [13]

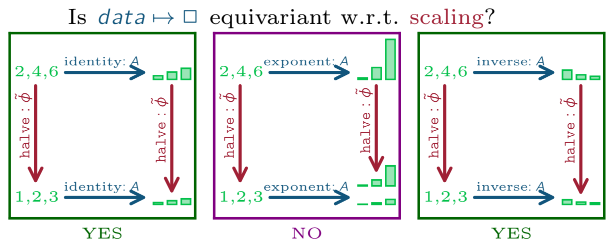

Figure 2: Encoding data as the bar height using an exponential transform is not equivariant because encoding the data and then scaling the bar heights yields a much taller graph then scaling the data and then encoding those heights using the same exponential transform function.

As illustrated in the commutative diagram in Figure 2, what this means is that the visual representation is consistent whether the data is scaled $(act(g, x))$ and then mapped (f) to a graphic or whether the data is mapped to a graphic that is then modified in a compatible way.

3.3 Homeomorphism

Definition 3.3. A function f is a *homeomorphism* if it is bijective, continuous, and has a continuous inverse function f^{-1}.

A function between topological spaces is continuous if the inverse image of any open sets is open. If two spaces are homeomorphic, they have identical topological properties and are considered topologically the same. Homeomorphism is almost always an unattainably strict notion of equality; therefore most topologists focus instead on weaker notions of equality, such as given by deformation retracts.

3.4 Deformation retraction

Definition 3.4. A *deformation retraction* of a space X onto a subspace A is a family of maps $f_t : X \to X, t \in I$ such that $f_0 = \nVdash$ (identity), $f_1(X) = A$, and $f_t|A = \nVdash$ for all t. The family f_t should be continuous in the sense that the associated map $X \times I \to X, (x, t) \mapsto f_t(x)$ is continuous. [9]

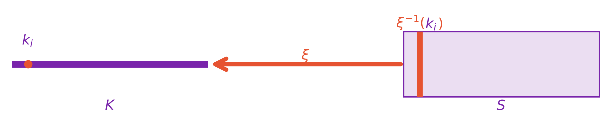

Figure 3: The deformation retraction ξ maps all the points in the orange band on the graphical base space $s \in S$ to a singular corresponding point k_i on the line K.

4 Fiber Bundles

Definition 4.1. A **fiber bundle** (E, K, π, F) is a structure with topological spaces E, F, K and bundle projection map $\pi : E \to K$ [15].

$$F \hookrightarrow E \xrightarrow{\pi} K \tag{1}$$

A continuous surjective map π is a **bundle projection** map when

1. all fibers $F_k = \pi^{-1}(k), k \in K$ are isomorphic to F. We call the uniquely determined F the fiber space, and it is implicitly encoded in the bundle projection since $F = \pi^{-1}(k)$ for any k.

2. each point k in the base space K has an open neighborhood U_k such that the total space E over the neighborhood is locally trivial.

Local triviality means $E|_U = U \times F$. In this paper we use $E|_U = \pi^{-1}(U)$ to denote the preimage of a neighborhood, and a **local trivialization** is a specific choice of neighborhoods and their preimages such that the fibers in each preimage are identical $F = F_k$ for all points $k \in U$.

Definition 4.2. A **section** $\tau : K \to E$ over a fiber bundle is a smooth right inverse of $\pi(\tau(k)) = k$ for all $k \in K$

$$\begin{tikzcd} F \arrow[r, hook] & E \arrow[d, two heads, "\pi"'] \\ & K \arrow[u, dashed, bend right, "\tau"'] \end{tikzcd} \tag{2}$$

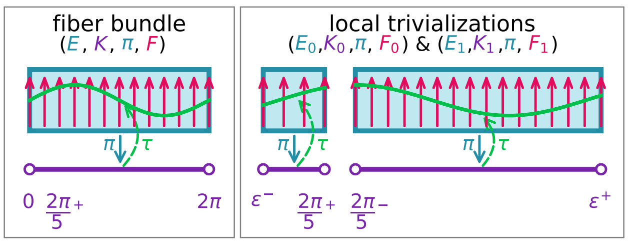

Figure 4: The space of all data values encoded by this fiber bundle can be modeled as a rectangle total space. Each dataset in this data space lies along the interval $[0, 2\pi]$ base space. Each dataset has values along the $-1 \to 1$ interval fiber. One dataset embedded in this total space is the sin section over the bundle.

For example, the fiber bundle in Figure 4 encodes the space of all continuous functions that have a domain of $[0, 2\pi]$ and range $[0, 1]$. Using a fiber bundle abstraction encodes that the dataset has a 1D linear continuity as the base space Kis the interval $[0, 2\pi]$ and a field type that is a `float` in the range $[0, 1]$. Therefore the type signature of the datasets in this fiber bundle, which is called a section τ, would be `dataset:` $[0, 2\pi] \to [0, 1]$. One such dataset (section) is the sin function, which as shown in Figure 4 is defined via a function τ from a point in the base space to a corresponding point in the fiber. Evaluating the section function over the entire base space yields the sin curve that is composed of points intersecting each fiber over the corresponding point. The local trivializations shown in Figure 4 are one way of decomposing the total bundle and conversely the bundle can be constructed from the local trivializations $K = K_0 \oplus K_1$. As shown, the section sin spans the trivializations in the same manner that it spans the bundle; this is analogous to how a dataset may span multiple tables or be collected in one table. The trivializations are glued together into the bundle at the overlapping region $\frac{2\pi}{5}$ by defining the transition map $F_1 \to F_2$. Because the fibers in Figure 4 at $\frac{2\pi}{5}$ are aligned, the transition map is an identity map that takes every value in F_1 and maps it to the same value in F_2 so that the sections, such as sin, remain continuous.

4.1 Trivial and non-trivial bundles

Generally, the distinguishing factor between a trivial bundle and a non-trivial bundle are how they are decomposed into local trivializations:

trivial bundle is directly isomorphic to $K \times F$. For any choice of cover of K by overlapping neighborhoods, we can choose local trivializations such that all transition maps are identity maps.

non-trivial bundle can not be constructed as $K \times F$. For any choice of local trivializations, there is at least one transition map that is not an identity [9].

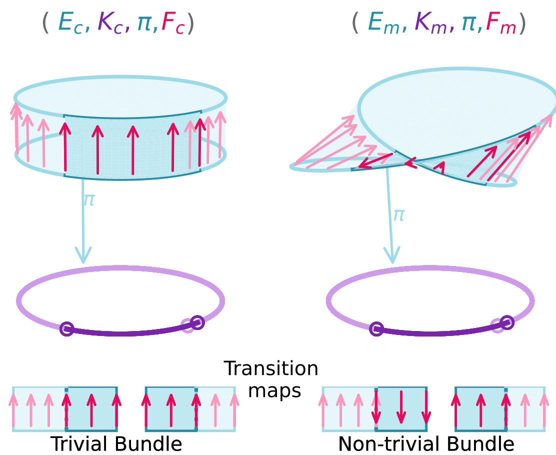

Figure 5: The cylinder is a trivial fiber bundle; therefore it can be decomposed into local trivializations that only need identity maps to glue the trivializations together. The mobius band is a non-trivial bundle; therefore it can only be decomposed into trivializations where at least one transition map is not an identity map.

In the example in Figure 5, we use arrows $\uparrow$ to denote fiber alignments. In the cylinder case the fibers all point in the same direction, which illustrates that they are equal $\uparrow = \uparrow$. In the Möbius band case, while the fibers in an arbitrary local trivialization are equal $\uparrow = \uparrow$, the fibers at the twist are unequal but isomorphic $\uparrow \cong \downarrow$. The cylinder and mobius band can be decomposed to the same local trivializations, for example the fiber bundles in Figure 4 In the cylinder case, the fibers in the overlapping regions of the trivializations are equal $F_0 \restriction_{U_1 \cap U_2} = F_1 \restriction_{U_1 \cap U_2}$; therefore the transition maps at both intersections map the values in the fiber to themselves $r \to r$. In the Möbius band case, while $F_0 \restriction_{(2\pi/5-\varepsilon, 2\pi/5+\varepsilon)} \to F_1 \restriction_{(2\pi/5-\varepsilon, 2\pi/5+\varepsilon)}$ can be chosen to be an identity map, the other transition map component $F_0 \restriction_{(-\varepsilon,\varepsilon)} \to F_1 \restriction_{(-\varepsilon,\varepsilon)}$ has to flip any section values. For example given $F_0 = \uparrow$ and $F_1 = \downarrow$, the transition map $r \mapsto -r$ maps each point from one fiber to the other $\uparrow \mapsto \downarrow$ such that any sections remain continuous even though the fibers point in opposite directions.

5 Presheaves and Sheaves

A common way of encapsulating a map from an arbitrary category to a category of sets is as a presheaf; Sheaves add on conditions for gluing individual sections over subspaces into cohesive sections over the whole space.

5.1 Presheaf

Definition 5.1. A **presheaf** $F : \mathcal{C}^{op} \to \mathbf{Set}$ is a contravariant functor from an object in an arbitrary category to an object in the category $\mathbf{Set}$[15]

$$\begin{tikzcd} \Gamma_{U_2}(E|_{U_2}) \arrow[r, dashed, "\hat{\phi}^*"] & \Gamma_{U_1}(E|_{U_1}) \\ U_2 \arrow[u, "\mathcal{O}_2"] & U_1 \arrow[l, "\hat{\phi}"'] \arrow[u, "\mathcal{O}_1"'] \end{tikzcd}$$

A functor is contravariant when the morphisms between the input objects go in the opposite direction from the morphisms between the output objects. The presheaf is contravariant when for every arbitrary morphism between input base spaces $\hat{\phi}: U_1 \to U_2$ there exists a corresponding pullback function between the sets of sections $\hat{\phi}^*: \Gamma_{U_2}(E \restriction_{U_2}) \to \Gamma_{U_1}(E \restriction_{U_{c1}})$.

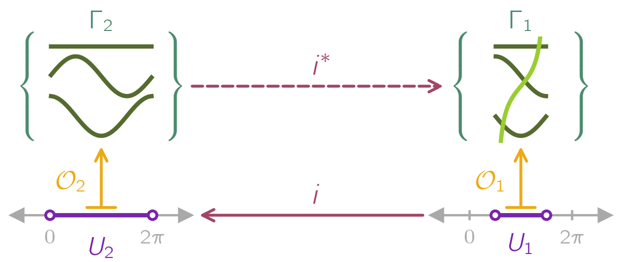

Figure 6: Modeling this data container as a presheaf specifies that since $\cos, \sin, \text{and} C$ are continuous over U_2, they must be continuous over U_1 since U_1 is a subset of and therefore must be included in U_2. Because $\tan$ is only defined over U_2, it does not need to be included in the set Γ_2.

For example in Figure 6, the transform $\hat{\phi}$ maps indexing points on the subspace U_1 to points on the larger subspace U_2. A constraint on presheaves is that because $\hat{\phi}$ exists, there must exist a corresponding pullback map $\hat{\phi}^*$ that changes the indexes of the data indexed by U_2 to the corresponding index in U_1. Since the $\mathtt{constant}, \mathtt{sin}, \mathtt{cos}$ functions are defined over the interval $[0, 2\pi]$, these functions must also be continuous over the sub-interval $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$; therefore the sections in Γ_2 must also be included in the set of sections over the subspace Γ_1. Sections (data) can also be present in subindexes without being present over the whole indexing space, as illustrated here by the presence of $\mathtt{tan}$ over U_1.

5.2 Sheaf

Definition 5.2. [1, 15] A **sheaf** is a presheaf that satisfies the following two axioms:

- *locality* two sections in a sheaf are equal $\tau^a = \tau^b$ when they evaluate to the same values $\tau^a|_{U_i} = \tau^b|_{U_i}$ over the open cover $\bigcup_{i \in I} U_i \subset U$ (indexed by I).

- *gluing* the union of sections defined over subspaces $\tau^i \in \Gamma(U_i, E|_{U_i})$ is equivalent to a section defined over the whole space $\tau|_{U_i} = \tau^i$ for all $i \in I$ if all pairs of sections agree on overlaps $\tau^i|_{U_i \cap U_j} = \tau^j|_{U_i \cap U_j}$

The gluing axiom says that a distributed representation of a dataset, which is a set of local sections, is equivalent to a section over the union of the neighborhoods of the local sections. The gluing axiom can also be used to generate the gluing rules used to construct non-trivial bundles from the set of trivial local sections. The locality axiom asserts that the glued section function is equivalent to a function over the union if they evaluate to the same values.

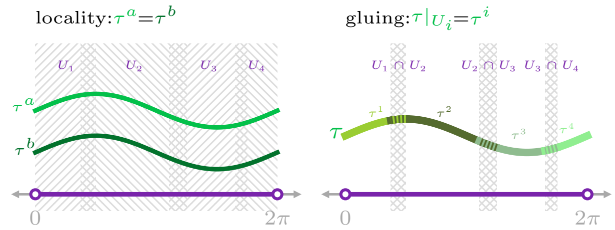

Figure 7: A sheaf has the conditions that sections are equal when they match on all subsets (*locality*) and that the sections can be concatenated when they match on overlaps *gluing*.

For example, in Figure 7, the τ^a and τ^b sin sections are equal because they match *locally* on all subsets. This is true whether sin is defined over parts ($\sin|_{U_i}$) or the whole space. If sin is defined over parts, then those parts can be *glued* together. The concatenated sin is continuous because the pieces of the section outside the overlap are continuous with the pieces inside the overlap. The glued sin is also equal to the non-glued sin because they match on the neighborhoods; therefore they are equivalent representations of the same section sin and so have the same mathematical properties.

5.3 Sheaf Transport Functors

Given a continuous function $\xi: S \to K$ between spaces S, K, we can construct functors that transport sheaves between the spaces [8].

Pushforward functor The pushforward (direct image) sheaf establishes which record returned by ρ corresponds to a point $k \in K$:

Definition 5.3. Given a sheaf $\mathcal{O}_{S,H}$ on S, the **pushforward** sheaf $\xi_* \mathcal{O}_{S,H}$ on K is defined as

$$\xi_*(\mathcal{O}_{S,H})(U) = \mathcal{O}_{S,H}(\xi^{-1}(U)) \tag{3}$$

for all opensets $U \subset K$[8].

The pushforward sheaf returns the set of sections ρ over K that corresponds to the space $\xi^{-1}(U) = W$. The pushforward functor ξ_* transports sheaves of sections on W over U

$$\Gamma_U(\xi_* H \restriction_U) \ni \xi_* \rho : U \to \xi_* H \restriction_U \tag{4}$$

such that it provides a way to look up which records returned by ρ correspond to k

$$\xi_* \rho(k) = \rho \restriction_{\xi^{-1}(k)} \tag{5}$$

such that $\xi_* \rho(k))(s) = \rho(s)$ for all $s \in \xi^{-1}(k)$.

Pullback functor The pullback (inverse image) sheaf establishes which record returned by τ corresponds to each point $s \in S$:

Definition 5.4. [8] Given a sheaf $\mathcal{O}_{K,E}$ on K, the **pullback** sheaf $\xi^* \mathcal{O}_{K,E}$ on S is defined as the sheaf associated to the presheaf $\xi^*(\mathcal{O}_{K,E})(W) = \mathcal{O}_{K,E}(\xi(W))$ for $\xi(W) \in K$.

The pullback sheaf returns the set of sections τ over the space S that corresponds to the space $\xi(W) = U$. The pullback ξ^* transports sheaves of sections on $U \subseteq K$ over $W \subseteq S$

$$\Gamma_W(\xi^* E \restriction_W) \ni \xi^* \tau : W \to \xi^* E \restriction_W \tag{6}$$

such that there is a way to then look up what records returned by τ correspond to a region s

$$\xi^* \tau(s) = \tau(\xi(s)) = \tau(k) \tag{7}$$

As ξ is surjective, there are many points $s \in W \subseteq S$ that correspond to a single point $\xi(s) = k$.

5.4 Artists over K and S

While Artists are defined as maps going from data over K to graphic over S they are compatible with maps defined solely over K or S. This is because when a sheaf is equipped with transport functors (ξ_*, ξ^*), maps between sheaves over one space $\mathrm{Hom}_{\mathcal{O}_K}$ are isomorphic to maps between sheaves over the other space $\mathrm{Hom}_{\mathcal{O}_S}$ [8]:

$$\begin{tikzcd}
\Gamma_U(\xi_* H\restriction_U) \arrow[from=r, "\xi_*"'] & \Gamma_W(H\restriction_W) \\
\Gamma_U(E\restriction_U) \arrow[u, "\mathrm{Hom}_{\mathcal{O}_K}"] \arrow[r, "\xi^*"'] \arrow[ur, "\mathrm{Hom}_{\mathcal{O}_K,\mathcal{O}_S}"] & \Gamma_W(\xi^* E\restriction_W) \arrow[u, "\mathrm{Hom}_{\mathcal{O}_S}"']
\end{tikzcd} \tag{8}$$

Both $\mathrm{Hom}_{\mathcal{O}_K}$ and $\mathrm{Hom}_{\mathcal{O}_S}$ are compatible, via composition with ξ, with a mapping from data sheaf in data space to graphic sheaf in graphic space $\mathrm{Hom}_{\mathcal{O}_K,\mathcal{O}_S}$. Artists are a subset of structure preserving functions in $\mathrm{Hom}_{\mathcal{O}_K,\mathcal{O}_S}$. The commutativity of Equation 8 means that an artist over data space $A_K : \tau \mapsto \xi_*\rho$, an artist over graphic space $A_S : \xi^*\tau \mapsto \rho$, and an artist $A : \tau \mapsto \rho$ are equivalent. Specifically, given Equation 7 (ξ^*) and Equation 5 (ξ_*), $A_K(\tau(k)) = A_S(\xi^*\tau(s)) = A(\tau(k)$ when $\xi(s) = k$.

5.4.1 Implementing the artist

Since the artist is a family of functions in the homset between sheaves, the isomorphism allows for the specification of the transformation from data as a combination of functions over different spaces:

$$\begin{tikzcd}
\Gamma_K(E) \arrow[r, "\nu^K"] \arrow[rr, bend left, "A^K"] \arrow[d, "\xi^*"'] \arrow[drr, "A"] & \Gamma_K(V) \arrow[r, "Q^K"] \arrow[d, "\xi^*"] \arrow[dr, "Q"] & \mathrm{Im}_A(K, \xi_* H) \\
\Gamma_S(\xi^* E) \arrow[r, "\nu^S"] \arrow[rr, bend right, "A^S"'] & \Gamma_S(\xi^* V) \arrow[r, "Q^S"] & \mathrm{Im}_A(S, H) \arrow[u, "\xi_*"']
\end{tikzcd} \tag{9}$$

This means that an artist over data space $A_K : \tau \mapsto \xi_*\rho$, an artist over graphic space $vartist_S : \xi^*\tau \mapsto \rho$, and an artist $A : \tau \mapsto \rho$ are equivalent such that:

$$\begin{aligned}
&\tau(k) = \xi^*\tau(s) \\
&\implies A_K(\tau(k)) = A_S(\xi^*\tau(s)) = A(\tau(k)) \\
&\implies \xi_*\rho(s) = \rho(s)
\end{aligned}$$

when $\xi(s) = k$.

6 Spivak's Type System

In Spivak's fiber bundle formulation of tables in relational databases [16, 17], Spivak introduces a *type specification* as a bundle map $\pi : \mathscr{U} \to \mathbf{DT}$. The base space $\mathbf{DT}$ is a set of data types $T \in \mathbf{DT}$ and the total space $\mathscr{U}$ is the disjoint union of the domains of each type $\mathscr{U} = \bigsqcup_{T \in \mathbf{DT}} \pi^{-1}(T)$ such that each element x in the domain $\pi^{-1}(T)$ is one possible value of an object of type T [17]. For example, if $T = \mathtt{int}$, then the image $\pi^{-1}(\mathtt{int}) = \mathbb{Z} \subset \mathscr{U}$ is the set of all integers and $x = 3 \in \mathbb{Z}$ is the value of one `int` object. Spivak's analogue of a database *schema* is the *simple schema of type* π which is defined as consisting of the pair (C, σ) where C is the set of field names and $\sigma : C \to \mathbf{DT}$ is a function from field name to field data type[17]. The function σ is composed with π such that $\pi^{-1}(\sigma(C)) \subseteq \mathscr{U}$; this composition induces a domain bundle $\pi_\sigma : \mathscr{U}_\sigma \to C$ that associates a field name $c \in C$ with its corresponding domain $\pi_\sigma^{-1}(C) \subseteq \mathscr{U}_\sigma$. Spivak defines a *record* as a function $r : C \to \mathscr{U}_\sigma$ from column name to type domain and the set of records on π_σ is denoted $\Gamma^\pi(\sigma)$[2]. Records must return an object of type $\sigma(C) \in \mathbf{DT}$ for each field $c \in C$. Spivak then describes tables as sections $\tau : K \to \Gamma^\pi(\sigma)$ from an indexing space K to the set of all possible records $\Gamma^\pi(\sigma)$ on the schema bundle, and his construction of a table generalizes to our notion of a dataset.

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[2] This paper follows the convention that Γ denotes a set.